Who benefits most?

Sorted effects, step by step · A colon cancer trial

Valentin Klotzbücher · Department of Clinical Research

Session overview

1. Is the effect the same for everybody?
2. Sorted effects, step by step: a colon cancer trial.
3. Can we trust the pattern?
4. Discussion.

Is the effect the same for everybody?

01

A trial reports one average effect. Whom does it describe?

One average, many patients

- A trial answers: does the treatment work **on average**?
- Patients differ, so the effect can differ too: large for some, small or none for others.
- So we also want to know: **who benefits most, and who least?**

“Why not … just estimate the distribution of effects?” *The Effect*, §22.3 (Huntington-Klein 2025)

Same average, different stories

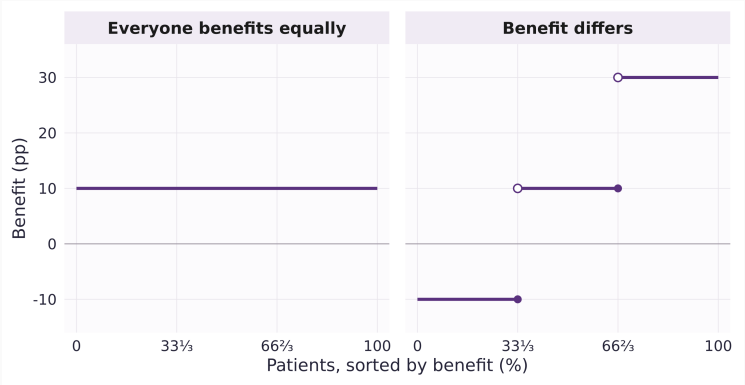
Three equally common kinds of patients, A, B and C. Without treatment, 40% of each die.

Benefit (pp)	A	B	C
Everyone benefits equally	+10	+10	+10
Benefit differs	-10	+10	+30

Both stories: average benefit **10 pp**. The average cannot tell them apart.

Constructed example with known risks, not data. Benefit = risk without treatment − risk with treatment.

Sorting tells them apart

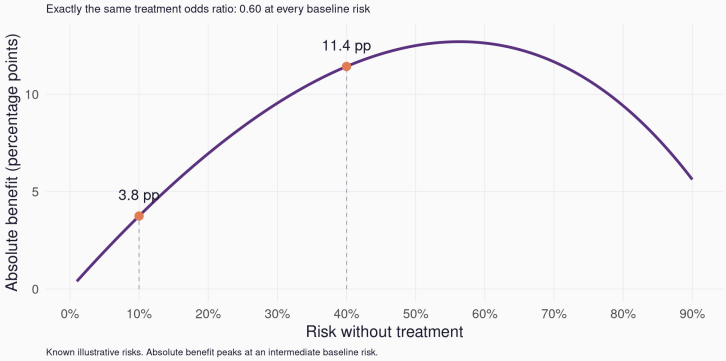


- Line patients up from least to most benefit.
- Everyone benefits equally: a flat line at **10 pp**.
- Benefit differs: a staircase from **−10 to +30 pp**.
- One third are worse off, despite a positive average.

In a trial we must estimate this curve. That is what sorted effects do.

Exact quantiles of the constructed example; no estimation.

Even a constant odds ratio gives different benefits



Algebraic illustration with known risks.

- Same odds ratio (**0.60**) for every patient.
- **10%** baseline risk → **3.8 pp**.
- **40%** baseline risk → **11.4 pp**.
- A logistic model already lets benefits differ; interactions let them differ more.

Sorted effects, step by step

02

A colon cancer trial: who benefits from chemotherapy?

The trial

- 619 patients with resected **stage III colon cancer**, randomized to observation or **levamisole + 5-FU**.
- Outcome: **death within three years**.
- Died: **34.7%** with observation (109/314), **25.7%** with chemotherapy (78/304).
- On average, about **9 fewer deaths per 100** treated patients. The same for everybody?

`survival::colon`, from the trial reports (Moertel et al. 1990, 1995). The third arm, levamisole alone, is not used. Analyses use the 593 patients with complete baseline data.

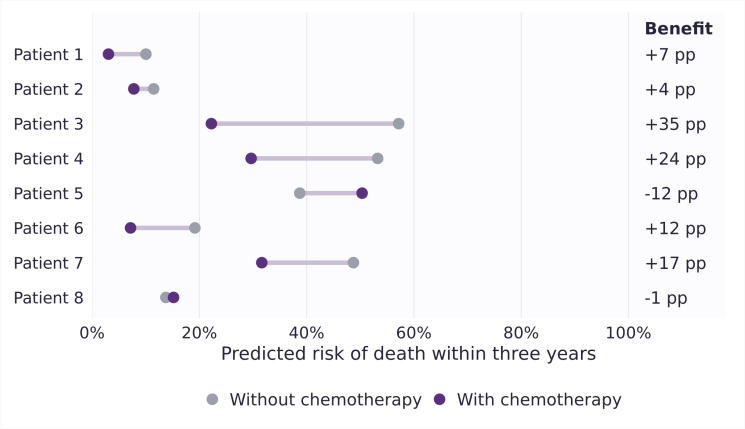
Step 1 · Let the effect differ

- A logistic model for death within three years.
- Chemotherapy's effect may change with **age, sex, number of positive nodes, tumour grade and tumour extent**.
- Bowel obstruction, tumour adherence to nearby organs and time from surgery to registration adjust the baseline risk.

Randomization makes the comparison fair; the model only describes how the effect varies.

Six treatment interactions, fixed before fitting; 19 parameters. Model code in backup.

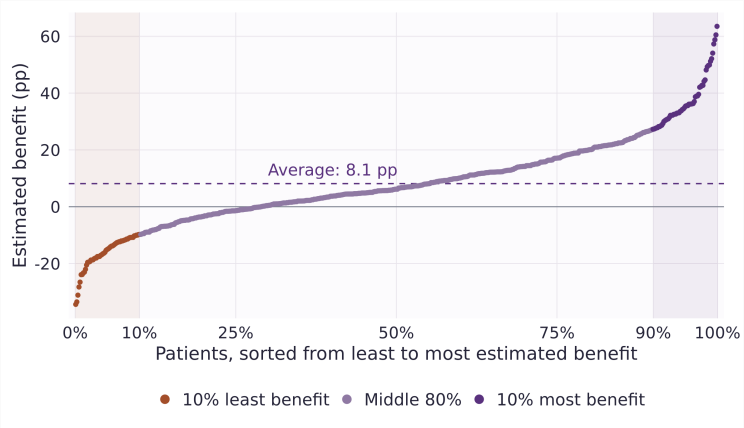
Step 2 · Predict both futures for every patient



- For each patient: predicted risk **without** and **with** chemotherapy.
- The gap is the **estimated benefit**.
- It is an average for patients like them, not a personal effect.

Eight of 593 patients, chosen to span the range and shown unsorted. Point estimates from the Step 1 model.

Step 3 · Sort them



- All 593 patients, from least to most estimated benefit.
- Average: **8.1 pp**. Middle 80%: **−10 to +27 pp**.
- Only **15%** of patients are within 3 pp of the average.
- The extremes are the least certain estimates.

Point estimates; uncertainty follows in chapter 03. In R: `SortedEffects::spe()` (call in backup).

Step 4 · Who is at each end?

Characteristic	10% least benefit	10% most benefit	Adjusted p
Women	100%	8%	.04
Women under 55	92%	0%	.04
Mean age (years)	43	53	.87
More than 4 positive nodes	13%	88%	.13
Poor differentiation	10%	42%	.95

- Least benefit: younger women with few positive nodes. Most: mostly men with many nodes.

This describes who sits at each end. It is not the effect of chemotherapy in women.

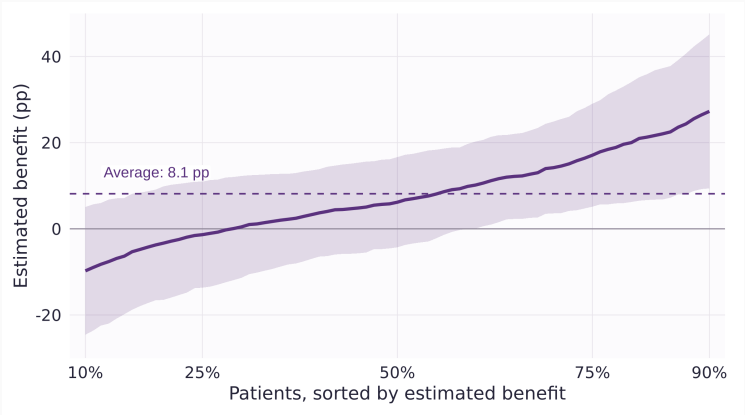
60 patients per tail. p-values are adjusted for comparing 12 traits. The package's own bootstrap broke down for these data, so they come from our checked re-run with stable starting values (backup).

Can we trust the pattern?

03

Is the spread real, or noise from one model and one trial?

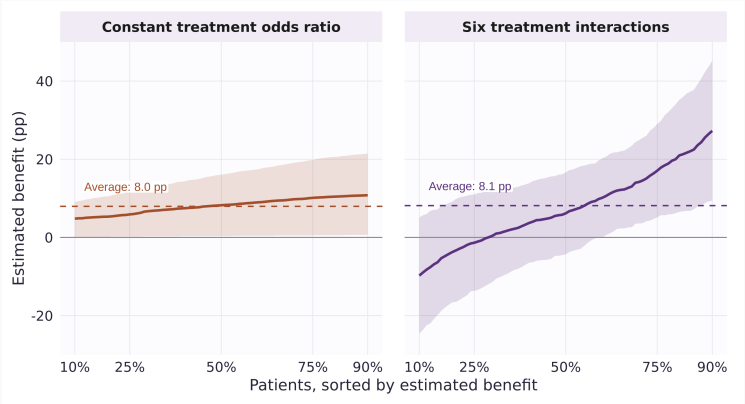
How sure are we?



- Shaded: a 95% band for the **whole curve**, not one point at a time.
- Near the bottom (10th percentile): **−9.8 pp**, band **−24.6 to +5.1**: we cannot tell whether these patients benefit.
- Average: **8.1 pp**, 95% CI **1.4 to 14.9**: clearly positive.

The average benefit is clear; the shape of the spread is much less certain.

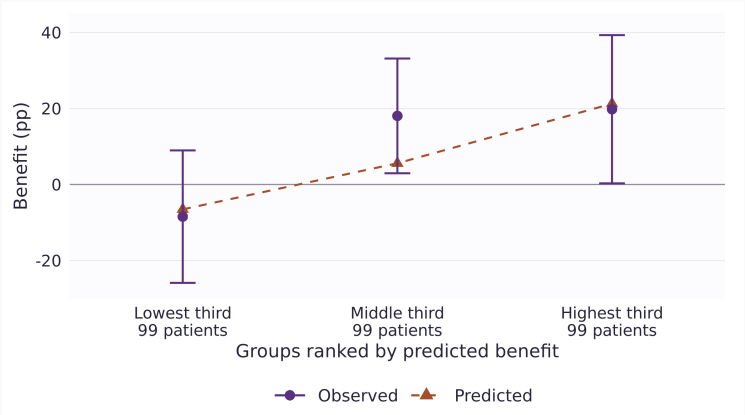
You find what your model allows



95% bands for the whole curve, one per model.

- Constant odds ratio: benefit varies only with baseline risk, a narrow spread.
- Six interactions: a much wider spread.
- Same average (**8.0 vs 8.1 pp**); the six interactions improve the fit (p = .014).
- More spread is not proof: noise spreads too.

Does it hold up in new patients?



- Fit on half the patients (296); rank the other half (297) blind to their outcomes.
- Higher-ranked patients did benefit more: slope **1.1** (1 = as predicted, 0 = no signal), p = .025.
- Lowest third: **−8.5 pp**, CI −25.9 to +9.0: again unclear.

One random split, balanced by treatment arm; three groups of 99; a 95% interval for each group. p adjusted for checking two models.

What could fool us?

- **Noise looks like spread:** 19 parameters for 179 deaths; 60 patients per tail.
- **Missing data:** keeping all 618 patients moves the p-value for the interactions from **.014 to .054**.
- **One split, one trial:** and colon was the third trial we tried, with no adjustment for that search.

A promising ranking to test in other trials, not a treatment rule.

Discussion

04

What would make sorted effects useful in our own trials?

What did we learn?

- The average (**8 pp**) hid a wide range of estimated benefits.
- Sorting turned a 19-parameter model into one curve and a table of who sits at each end.
- The pattern is uncertain and depends on the model; it needs other trials.

Which result would convince you that the pattern is real?

Questions for discussion

- **Spread or noise?** 29% of patients get a negative estimated benefit. Would you shrink the interactions first?
- **Planning:** would you prespecify sorted effects in a trial's analysis plan?
- **Risk or interactions:** start from baseline risk, or let the effect depend on traits?
- **Scale:** absolute or relative benefit?

Next time: causal forests

- **Here:** we chose which interactions to allow.
- **Causal forests:** let the data choose. A tree splits patients where the effect differs most, say under and over 65; a forest averages many trees.
- **Same caveat:** neither method creates causality. Both need a design, here randomization.

The Effect, §22.3.1 (Huntington-Klein 2025); generalized random forests (Athey, Tibshirani, and Wager 2019). No forest is fitted in this session.

Reading and materials

- **Intuition:** *The Effect*, chapter 10 (treatment effects) and §22.3 (heterogeneous effects), free online.
- **Tutorial:** the SortedEffects R package (Chen et al. 2020).
- **Method:** the sorted effects paper (Chernozhukov, Fernández-Val, and Luo 2018).
- **Code, report and slides:** github.com/valentinklotzbuecher/jcod.

Additional material

+

Code, bootstrap details, full tables and assumptions.

Backup · The estimand

$$\Delta(x) = \mathsf{Pr}(Y = 1 \mid A = 0, X = x) - \mathsf{Pr}(Y = 1 \mid A = 1, X = x)$$

$$\mathsf{SPE}(u) = Q_{\Delta(X)}(u), \qquad \mathsf{APE} = E[\Delta(X)]$$

- Y: death within three years; $A = 1$: levamisole + 5-FU, $A = 0$: observation; x: the eight baseline characteristics in the model.
- With randomization, $\Delta(x) = E[Y(0) - Y(1) \mid X = x]$: an average over patients with profile x, not an individual effect.
- The distribution of X, here the 593 observed profiles, defines the population summarized.
- SPE(u), the sorted partial effect, is the u-th percentile of the effects; APE is their average.
- A quantile of effects differs from a difference of outcome quantiles.

Backup · The model in R

```
1  fm <- outcome ~ treatment + splines::ns(age, df = 3) + female +
2    age_z:female + log_nodes + poor_diff + extensive +
3    obstruct + adhere + surg_late +
4    I((treatment == "chemotherapy") * age_z) +
5    I((treatment == "chemotherapy") * female) +
6    I((treatment == "chemotherapy") * age_z * female) +
7    I((treatment == "chemotherapy") * log_nodes) +
8    I((treatment == "chemotherapy") * poor_diff) +
9    I((treatment == "chemotherapy") * extensive)
10
11 fit <- glm(fm, data = dat, family = binomial())
```

1–3: common model · **4–9:** six treatment interactions · **11:** fit the model.

Comparison model: omit lines 4–9. `age_z = (age - 60) / 10`; `log_nodes = log1p(nodes)`.

Backup · Predict both conditions in R

```r
control <- treated <- dat
control$treatment <- "observation"
treated$treatment <- "chemotherapy"

benefit <- predict(fit, control, type = "response") -
  predict(fit, treated, type = "response")
```

- Keep every patient's baseline characteristics fixed.
- Change only treatment assignment.
- Subtract predicted **risks**, not log odds.

Backup · Sorted effects with spe()

```r
spe_result <- SortedEffects::spe(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  us = seq(.10, .90, .01),
  b = 499,
  alpha = .05, bc = FALSE,
  seed = 420
)
```

Fit the outcome model
Our chosen formula and a logistic link.
Set the contrast
This ordering gives observation − chemotherapy.
Choose the percentiles
10th–90th, in one-percentile steps.
Quantify uncertainty
499 resamples; 95% band; no bias correction.

Uniform coverage concerns the chosen percentile grid and requires valid effect estimation and regularity conditions.

Backup · How is the uniform band built?

1. Refit in each of **499** bootstrap draws and re-sort the fitted benefits at 81 percentiles, 10th–90th.
2. Scale each percentile by its bootstrap interquartile range ÷ 1.349.
3. Critical value: 95th percentile of the **largest** scaled deviation across all 81 percentiles.
4. Band: estimate ± critical value × scale; endpoints rearranged to stay monotone.

Model	Uniform critical value	Pointwise
Six treatment interactions	2.79	1.96
Constant treatment odds ratio	2.17	1.96

The package's own bootstrap (SortedEffects 1.7.0), which we reproduced independently; critical values read off the saved bands. Rearrangement: Chernozhukov, Fernández-Val, and Galichon (2009).

Backup · Comparing the tails with ca()

```r
ca_result <- SortedEffects::ca(
  fm = fm, data = dat,
  method = "logit",
  var_type = "categorical", var = "treatment",
  compare = c("chemotherapy", "observation"),
  u = .10,
  interest = "moment", cl = "diff",
  t = colon_characteristics()$variable,
  b = 499, alpha = .05,
  bc = FALSE, seed = 420
)
```

Model and contrast
Same mortality model; benefit = observation − chemotherapy.
Choose the tails
Bottom and top 10%: 60 patients each here.
Compare characteristics
High − low means, across all twelve traits.
Bootstrap inference
499 resamples; SEs and p-values adjusted across the twelve traits.

`ca()` is classification analysis. For these data its own bootstrap breaks down; the p-values we show come from our checked re-run with stable starting values (next slide).

Backup · Why the package's tail bootstrap failed here

- In each bootstrap resample, `ca()` refits the logistic model with weights scaled to sum to **20,000**.
- On these data, **141 of 499** refits did not converge, and **137** resamples gave undefined tail means.
- The function still printed normal-looking SEs and p-values. We do not use them.
- Our re-run starts each refit from stable values: **all 499** refits converge, and none are dropped.

Check every bootstrap refit: output that looks normal can still be wrong.

Same resamples, model, tail definitions and tail means; only the numerical starting point changed. Details in the colon report.

Backup · Calculate tail means

```r
cutoff <- Hmisc::wtd.quantile(
  benefit, rep(1, nrow(dat)), c(.10, .90)
)
tails <- list(low = benefit <= cutoff[1],
              high = benefit >= cutoff[2])
traits <- colon_characteristics()$variable
lapply(tails, function(i)
  colMeans(dat[i, traits]))
```

- Select tails by fitted benefit, then summarize baseline characteristics.
- Both tails can contain patients from either randomized arm.

Descriptive calculation only. `R/colon-walkthrough.R` verifies these means against the saved classification table.

Backup · Classification: demographics and nodes

Characteristic	Low	High	Difference	SE	Joint p
Age (years)	43.2	52.9	+9.7	7.1	0.872
Female (%)	100.0	8.3	-91.7	21.4	0.042
Men aged 65+ (%)	0.0	6.7	+6.7	14.2	1.000
Women under 55 (%)	91.7	0.0	-91.7	21.1	0.040
Positive nodes	2.6	8.4	+5.8	2.2	0.297
More than 4 nodes (%)	13.3	88.3	+75.0	22.1	0.132

Difference = high − low. Percentages differ in percentage points.

60 patients per tail. SEs and p-values from our checked re-run of the tail bootstrap (stable starting values, 499 resamples); p-values adjusted across all twelve traits, not just these six.

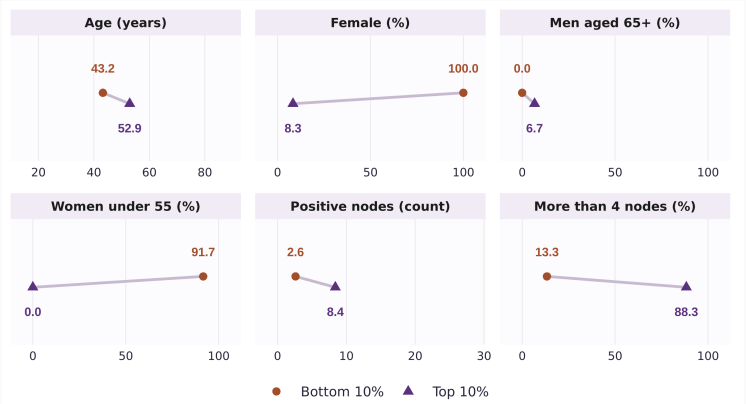
Backup · Classification: clinical characteristics

Characteristic	Low	High	Difference	SE	Joint p
Poor differentiation (%)	10.0	41.7	+31.7	27.7	0.948
Serosa or beyond (%)	98.3	85.0	-13.3	17.4	0.992
Bowel obstruction (%)	35.0	20.0	-15.0	9.6	0.762
Perforation (%)	3.3	3.3	+0.0	3.7	1.000
Tumour adherence (%)	16.7	15.0	-1.7	10.5	1.000
Long surgery interval (%)	33.3	25.0	-8.3	10.8	0.990

Difference = high − low, in percentage points.

“Long surgery interval”: surgery-to-registration category. SEs and p-values from our checked re-run of the tail bootstrap, adjusted across all twelve traits.

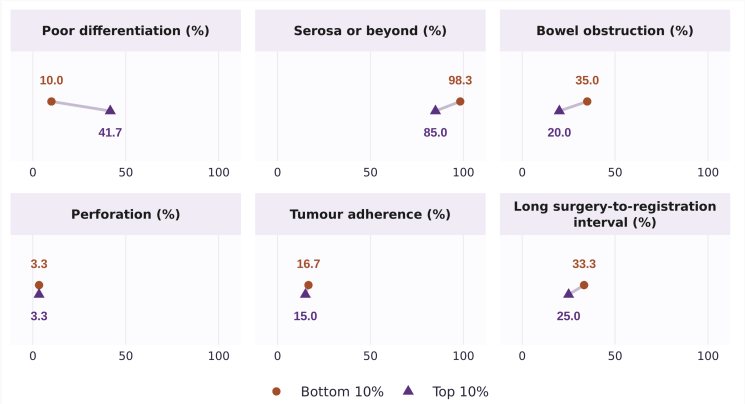
Backup · Tail profiles: age, sex and nodes



- Low tail: **all women**, mean age **43**.
- High tail: **92% men**, mean age **53**.
- More than four nodes: **13% vs 88%**.
- Men aged 65+: **7% of the high tail**.

Descriptive means; 60 patients per tail. Connecting segments are not confidence intervals.

Backup · Tail profiles: clinical characteristics



“Adherence” denotes tumour adherence to nearby organs.

- Poor differentiation: **10% vs 42%**.
- Bowel obstruction: **35% vs 20%**.
- Perforation: **3% in both tails**.
- Adjusted p-values: classification tables.

Backup · What is resampled?

- `spe()`**:** refits the model in each resample, then sorts the predicted effects for the original 593 patients.
- `ca()`**:** refits the model in each resample, recomputes who falls in each tail, and recomputes the tail means.
- **Held-out check:** fits on one half only, freezes the ranking, then looks at outcomes in the other half.

Backup · Cohort and sensitivity checks

Analysis	Patients	Average benefit	Interaction p
Complete baseline and endpoint	593	8.1 pp	.014
Missing baseline values with indicators	618	8.4 pp	.054
Early-censored outcome assigned 0	594	8.0 pp	.017
Early-censored outcome assigned 1	594	8.4 pp	.010

- The latter two retain complete baseline variables.
- Eight records disagree on the supplied vs count-derived four-node indicator; the report includes a coding check.

Backup · When is the heterogeneity causal?

- **Identification, both:** randomization or no unmeasured confounding; overlap; consistency/no interference; appropriate handling of selection and missing data.
- **Regression → sorted effects and tail comparison:** reliable conditional-risk estimates; suitable convergence and regularity of the effect distribution.
- **Forests:** enough local data, regularity, suitable nuisance estimates and honest fitting.
- **Inference:** pointwise forest intervals do not automatically cover sorted effects or selected tails.

Both concern conditional average effects. Fitted spread alone does not establish true heterogeneity. (Chernozhukov, Fernández-Val, and Luo 2018; Athey, Tibshirani, and Wager 2019)

Backup · Where do sorted effects fit?

Approach	Question	Watch for
Subgroup forest plot	Does the effect differ by one trait?	One trait at a time; many tests
Risk modeling	Does absolute benefit change with baseline risk?	Misses modifiers unrelated to risk
Effect modeling	Which traits modify the effect?	Overfitting; needs validation
Sorted effects + classification	How are fitted effects distributed, and who is in the tails?	Only as good as the effect estimates

Here, the constant-odds-ratio model is risk-based; the six interactions are effect modeling.

Risk and effect modeling as defined in the PATH statement (Kent et al. 2020).

Backup · How a forest builds a comparison

For a patient profile, which treated and control patients are comparable?

1 · Grow a tree

Split baseline profiles to find groups with different treatment–control contrasts.

2 · Grow many trees

Different samples and candidate splits give different groups of similar patients.

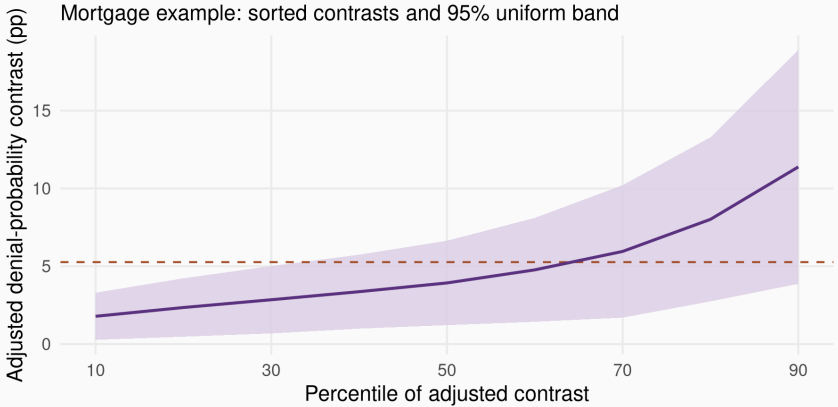
3 · Combine the comparisons

Patients repeatedly grouped with the target profile get more weight in its treatment comparison.

Many trees reduce dependence on any one noisy split.

“Honest” fitting uses separate observations to choose splits and estimate effects. Randomized-trial intuition; no forest was fitted here. (Athey, Tibshirani, and Wager 2019)

Backup · Original published example



Tutorial benchmark: adjusted observational disparity (Munnell et al. 1996; Chen et al. 2020).

References I

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